HDOC Day-28 - Solution

Lakshya Yadav | Student ID: 590034429

Problem / Objective

Read positive row and column dimensions, create a variable-length 2D integer array, accept elements column-by-column, print them in row-major order using a single linear index, then copy the 2D array to a 1D array in row-major order and display both arrays.

Algorithm

Repeatedly read rows and columns until both are positive. Declare VLA a[rows][cols]. For column-wise input, make the outer loop traverse columns and the inner loop traverse rows. For one-index row-major output, map linear index k to row k/cols and column k%cols. Finally copy elements row-by-row into a 1D VLA.

C Program

```
#include <stdio.h>
int main(void){
int r,c,i,j;
do{
printf("Rows and columns (>0): ");
scanf("%d%d",&r,&c);
}
while(r<=0||c<=0);
int a[r][c],b[r*c],k=0;
printf("Enter %d elements column-wise:\n",r*c);
for(j=0;
j<c;
j++)for(i=0;
i<r;
i++)scanf("%d",&a[i][j]);
printf("Row-major using one linear index:\n");
for(k=0;
k<r*c;
k++){
printf("%d ",a[k/c][k%c]);
if((k+1)%c==0)printf("\n");
}
k=0;
for(i=0;
i<r;
i++)for(j=0;
j<c;
j++)b[k++]=a[i][j];
printf("2D array:\n");
for(i=0;
i<r;
i++){
for(j=0;
j<c;
j++)printf("%d ",a[i][j]);
printf("\n");
}
printf("1D copy: ");
for(i=0;
i<r*c;
i++)printf("%d ",b[i]);
printf("\n");
return 0;
}
```

Test Cases / Expected Results

Case	Input	Expected	Status
1	r=2,c=3; column input 1 4 2 5 3 6	2D rows become 1 2 3 and 4 5 6	Pass

Case	Input	Expected	Status
2	r=0,c=3 then r=2,c=2	Dimensions requested again	Pass
3	2x2: 10 30 20 40	1D row-major copy: 10 20 30 40	Pass

Result / Notes

The expressions k/cols and k%cols convert a single linear index into the correct row and column positions.